BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: 2461 / 2468

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q1. (Marks 17) Q. 1

Q1. Milk and milk products are preserved by

(A) (B)

(C) (D)

Q2. In alanine R is

(A) (B)

(C) (D)

Q3. Competitive inhibitor of succinic acid is

(A) (B)

(C) (D)

Q4. Funciton of Golgi apparatus is concerned with

(A) (B)

(C) (D)

Q5. Inflammation of Liver is called

(A) (B)

(C) (D)

Q6. Bacteria grows rapidly in this phase

(A) (B)

(C) (D)

Q7. Plasmodium reproduces asexually in

(A) (B)

(C) (D)

Q8. Major structural component of fungus cell wall is

(A) (B)

(C) (D)

Q9. The structure that includes all others is

(A) (B)

(C) (D)

Q10. The sponge that is called venus flower basket

(A) (B)

(C) (D)

Q11. Animals having compound eyes are

(A) (B)

(C) (D)

Q12. The other name of Calvin cycle is

(A) (B)

(C) (D)

Q13. The most abundant chlorophyll is

(A) (B)

(C) (D)

Q14. Serum electrolyte imbalance occurs in

(A) (B)

(C) (D)

Q15. Respiratory pigment in humans is

(A) (B)

(C) (D)

Q16. Excess fluid in the tissue of the body is

(A) (B)

(C) (D)

Q17. Narrowing and hardening of arteries is called

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 16) 2. Answer briefly any Eight parts from the followings:-

(i) What is an ester? Express it with an equation.

(ii) How does substrate concentration affect the reaction rate of enzyme?

(iii) Differentiate binding and catalytic site of enzyme.

(iv) Write down any two characteristics of enzyme.

(v) How can fungi grow on fruits even in refrigerator?

(vi) Differentiate obligate and facultative parasitic fungi.

(vii) What is marsupium? Give its functions.

(viii) Write down the economic losses caused by phylum mollusca.

(ix) What are the symptoms of disease caused by hook worms?

(x) Why notochord is important in chordates?

(xi) What are the accessory pigments in plants? Give an example.

(xii) How glucose is prepared for production of energy?

Q3. (Marks 16) 3. Answer briefly any Eight parts from the followings:-

(i) What are bio-pesticides? Give example.

(ii) Define bioremediation and endangered species.

(iii) Differentiate chromoplast and leucoplast.

(iv) Give role and composition of cytoskeleton.

(v) Write down the importance of algae.

(vi) Why the euglenoids are placed in algae as well as in protozoan.

(vii) Why Kindom Protista is regarded as a polyphyletic group of organisms?

(viii) Write two characteristics of apicomplexans.

(ix) Mention two changes in chest cavity that cause expiration.

(x) Differentiate the bronchi and bronchioles.

(xi) What are lymph nodes? Give their function.

(xii) How systolic pressure differs from diastolic pressure?

Q4. (Marks 12) 4. Answer briefly any Six parts from the followings:-

(i) Define reverse transcriptase enzyme. Also give its function.

(ii) Differentiate Lag phase and log phase.

(iii) Give the scientific name of tomato.

(iv) How microsporophylls are different from megasporophylls?

(v) Draw the labeled diagram of prothallus (Adiantum)

(vi) Write the names of four extinct earliest vascular plants.

(vii) What is gastrovascular cavity? Give an example.

(viii) Draw labelled diagram of large intestine of man.

(ix) What do you know about "Hunger Pangs".

Q. Section ---------- II

Q5. Note: Attempt any three questions.

(a) Describe role of biology in protection and conservation of environment.

(b) What is photorespiration? Give its consequences.

(a) Write an essay on Acylglycerols.

(b) Discuss economic gains due to fungi.

(a) What is the chemical composition of plasma membrane? Discuss the structure of plasma membrane.

(b) Discuss/write a note on two methods of nutrition in plants.

(a) Explain symptoms, causes, spread and preventions of the AIDS.

(b) "Transpiration is a necessary evil" justify the statement.

(a) Describe different physical and chemical methods to control bacteria.

(b) Give the diagrammatic representation of non-cyclic electron flow in photosynthesis.

GROUP – II

Q1. (Marks 17) Q. 1

Q1. $CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow HCO_3^- + H^+$ An above reaction occurs at:

(A) (B)

(C) (D)

Q2. If cell has + 600 kPa pressure potential (Ψp) and -800 kPa water potential (Ψw). Then what would be its osmotic potential (Ψs).

(A) (B)

(C) (D)

Q3. Which of the following cells function as multisensory hydraulic valves?

(A) (B)

(C) (D)

Q4. In which of the following era, Mammals became dominent:

(A) (B)

(C) (D)

Q5. How many water molecules will be released to produce tripeptides?

(A) (B)

(C) (D)

Q6. Activated enzyme consisting of polypeptide chain and a co-factor is known as;

(A) (B)

(C) (D)

Q7. A crista is chemically composed of:

(A) (B)

(C) (D)

Q8. Influenza virus is:

(A) (B)

(C) (D)

Q9. Discoloration of teeth is caused by

(A) (B)

(C) (D)

Q10. All Algae form flagellated motile cells in their life cycle except:

(A) (B)

(C) (D)

Q11. Fungi have chitin, a chemical found in external skeleton of:

(A) (B)

(C) (D)

Q12. An egg containing female gametophyte in plants is called:

(A) (B)

(C) (D)

Q13. Members of which phylum are asymmetrical?

(A) (B)

(C) (D)

Q14. Which of the following vertebrates are hermephrodite?

(A) (B)

(C) (D)

Q15. In Alcoholic fermentation, Acetaldehyde is reduced to ethanol by:

(A) (B)

(C) (D)

Q16. How many number of ATP molecules are required to generate one mole of Triose sugar?

(A) (B)

(C) (D)

Q17. Deficiency of which element causes stunted growth of roots?

(A) (B)

(C) (D)

Q. Section ---------- I

Q2. (Marks 16) 2. Answer briefly any Eight parts from the followings:-

(i) Differentiate between nucleotide and nucleoside.

(ii) What is induced fit model? Who proposed it?

(iii) Why pepsin is produced in its inactive form called pepsinogen?

(iv) Define enzyme-substrate complex.

(v) Give names of four plant diseases caused by fungi.

(vi) Name key mutulistic symbiotic associations of fungi.

(vii) Differentiate the Ostia and Osculum.

(viii) Give two fundamental characters of chordates.

(ix) Give the function of mantle and redula in mollusks.

(x) What are running birds? Quote two examples.

(xi) What is Rubisco? Give its function?

(xii) What are accessory pigments? State their role.

Q3. (Marks 16) 3. Answer briefly any Eight parts from the followings:-

(i) What is Bioremediation? Give one example.

(ii) What is meant by Integrated disease management?

(iii) Differentiate the phagocytosis and Pinocytosis.

(iv) What are storage diseases? Give two examples.

(v) How Algae (Plant-Like Protists) differ from plants?

(vi) Write down importance of Chlorella.

(vii) What are Choanoflagellates? Give their evolutionary link with sponges.

(viii) Basically the kingdom Protista is defined by exclusion. How?

(ix) Write four properties of Respiratory Surface.

(x) How Respiratory Distress syndrome is caused?

(xi) Differentiate the Thrombous formation and embolus.

(xii) How vasodilation and vasoconstriction regulate blood flow?

Q4. (Marks 12) 4. Answer briefly any Six parts from the followings:-

(i) Fungi and animals are heterotrophs but place in separate Kingdoms. Why?

(ii) What do you know about bacilli bacteria? Give an example.

(iii) What is the importance of alternation of generation?

(iv) Why anthoceropsida is considered advanced than any other bryophytes?

(v) What is the difference between Monocots and Dicots?

(vi) Compare homospory and heterospory?

(vii) What are deficiency symptoms of Magnesium and Phosphorus in plants?

(viii) What are obligate and facultative parasites?

(ix) What is the cause of diarrhoea and constipation?

Q. Section ---------- II

Q5. Note: Attempt any three questions.

(a) What is cloning? Write down two different methods of cloning.

(b) Discuss the factors which affect the oxygen carrying capacity of haemoglobin.

(a) What is RNA? Give its three types with their role.

(b) Why the fungi were placed in a separate kingdom i.e., Kingdom Fungi? Explain.

(a) Compare prokaryotic with Eukaryotic cell.

(b) Write the structure and functions of stomach by drawing its labeled sketch.

(a) Discuss Hepatitis in detail.

(b) Describe cohesion-tension theory of water movement in xylem, which supplies cohesion and what is the source of tension? How does these forces interact to move water through plants.

(a) Describe Nutrition in bacteria.

(b) What is chemiosmosis? Describe cyclic phosphorylation by sketching it.